Solar Forecasting using Historical Sunspot Data

CSC-475: Seminar in Computer Science

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Abstract

Solar weather has been around as long as time itself and was first related to auroral activity in the 1750s, and later associated with events such as the first observed solar flare in the 1850s. With the rise of modern technology and reliance on space based satellite systems, solar weather is becoming increasingly important, primarily due to the ability that it has to disrupt and disable these systems. Solar weather also imposes operational effects as seen most often in low latitude and high latitude regions. These effects include GPS signal scintillation (signal interference and delays) and high frequency radio disturbances from 3 - 30 MHz bands. Currently, Solar forecasting with great accuracy far in advance is nearly impossible, and is only effective when the CME has already formed on the Sun's Corona, meaning that it is within 15-72 hours out from impact with Earth's magnetic sphere. With recent advances in machine learning and artificial intelligence based systems, interesting insights provide a hopeful future in solar weather forecasting that will prevent the human race from being sent back to the stone age from another Carrington strength event. In this project, I used historic sunspot data starting in 1818 in combination with the length of the average solar cycle and utilized the ARIMA python library and SARIMAX algorithm to predict the general number of sunspots per day for the rest of the current solar cycle, which directly related to solar interference.

1 Introduction

Solar weather forecasting has been at the forefront of technological advancement for the last several years, and increasingly so since the early 2010s. In 2002, during Operation Anaconda in the Paktia Province of Afghanistan, a low latitude region, a MH-47 Chinook carrying soldiers was shot down due to a communication failure regarding a "hot" landing zone. Over ten years later, scientists at Johns Hopkins University Applied Physics Laboratory (APL) discovered that there had been a plasma bubble overhead at the time of the operation, revealed through footage captured using a Global Ultraviolet Imager (GUVI) attached to a NASA spacecraft [1].

This is just one example of how scintillation has had a detrimental impact on operations, and will certainly play a role in the United States' Arctic Strategy, which concerns National Security in and around Arctic Regions. Combating scintillation will be increasingly important due to the melting of arctic ice caps, which will in turn open up new shipping routes and will make the Arctic Ocean sailable, for friendly and adversarial countries.

This problem has been the interest of many scientists working very quietly at Johns Hopkins University Applied Physics Lab (APL) in Laurel, Maryland for the last 10 years, and will continue to be a primary concern for the foreseeable future. While these scientists are developing projects such as DAGGER for NASA, these projects are often inaccessible and protected by layers of "TOP SECRET" and "SENSITIVE COMPARTMENTALIZED INFORMATION" (TS/SCI), which not only requires the appropriate clearance, but also a SCIF and need to know (NK), which almost nobody would be able to obtain for these types of projects other than those developing them.

This report deals with the issue of predicting the solar cycle and later solar weather using the seasonal autoregressive model, also known as SARIMAX. The ultimate goal of this project is to identify the parameters needed for a decent prediction using the historical number of sunspots throughout solar cycles 1-24, and then use these results to forecast solar cycle 25. After this, I would like to incorporate Kp and ap data in some type of forecast for the chance of experiencing geomagnetic interference.

This project will utilize the Solar Influences Data Analysis Center's (SIDC) publicized historical sunspot record, recorded since 1818 and daily since the early 1900s. For the Kp and ap indices, I will use Geomagnetic Observatory Niemegk, GFZ German Research Centre for Geosciences's publicized Kp and ap records, recorded 3 hourly since 1932.

Using these datasets, an ideal prediction would provide a projected number of sunspots, and based on the number of sunspots the number of notable solar events of concerning strength for a certain month and year, and a projected Kp for a given day, which would be based on historical trends along with the projected sunspot number. While this is a lofty goal, I will continue working on this project well into the future in order to advance my knowledge and create software that is free and available for anyone to use.

Based on the current results, I am predicting a weaker solar cycle, with less sunspots than average. I am predicting the next solar minimum in 2029, which is on par with trends and published predictions. While my model can predict years in advance, predicting so far in advance without previous years actual data proves unreliable and tends to skew the trend.

2 Related Work

While not required, solar weather forecasting is a lot easier to digest with a very thorough understanding of the underlying physics. The aforementioned sunset dataset, updated daily, can be referenced at the link included in the references. I will discuss this dataset more in the next section [2]. Similarly to the sunspot dataset, I will also discuss in detail the Kp and ap 3 hourly dataset in the next section as well [3]. Due to the raw nature of solar physics and plasma physics, any project dealing with these topics should include a heavy dose of literature review in the topics of plasma physics, magnetic reconnection, charged particle behavior in the presence of a magnetic field (electromagnetic theory and electrodynamics), the nature of geomagnetic events such as geomagnetic storms and geomagnetic substorms, and the anatomy of Earth's magnetosphere in the presence of a strong geomagnetic event. A project such as this should also include computer science related literature review of the ARIMA library and machine learning, which will be discussed after the physics foundation is laid.

The solar cycle is the driving force of all solar weather. The solar cycle is a 11 year cycle, meaning there is typically 11 years from one solar minimum to another solar minimum. Solar minimum refers to a time in which the Sun is firing off less Coronal Mass Ejections, and the ejections that do happen are typically pretty weak. Solar maximum is exactly the opposite of this, meaning the Sun is very active, and is very often firing off strong CMEs [4]. Solar maximum is typically when most satellites and power grids on Earth experience irreparable damage.

First, one needs to consider a "satellite view" of solar weather. When thinking about solar weather, think about the Sun as a cannon. When this cannon fires a projectile, there is a muzzle flash, just as there would be when a cartoon cannon fires. The "muzzle flash" of this cannon is what a solar flare is, which is generally unimportant for our purposes. On Earth, this "muzzle flash" is observed within 8 minutes of a solar weather event, as it travels at the speed of light. The projectile that is fired is what is called a Coronal Mass Ejection or CME. A CME is one of the consequences of living near almost any star, but most commonly Red Giants, Red Dwarfs, or Yellow Dwarfs, like our Sun [4]. CMEs, as the name indicates, is solar mass being ejected from the sun to the Earth. This mass comes in the form of one to ten billion tons of charged particles, and can travel several million miles per hour, reaching Earth anywhere from 15-72 hours.

Next, the solar mass from this CME is going to reach Earth. After reaching Earth, these charged particles can do several things, but for the sake of brevity I will only be discussing the most common two cases. One case is that they combine with the outer neutral layers of the magnetosphere, and get stuck in a donut shaped region called the "ring current" of Earth. This first phenomenon is called a geomagnetic storm, and takes on average 2-4 days to diffuse to a normal level of interference [5]. Geomagnetic storms happen on average 1-2 times per month at the most active period of a solar cycle. As shown in the image below, geomagnetic storms usually have a somewhat sudden onset, followed a buildup of charged particles in the ring current (lowest point on the chart), and are then in turn diffused over several days due to particle collisions.

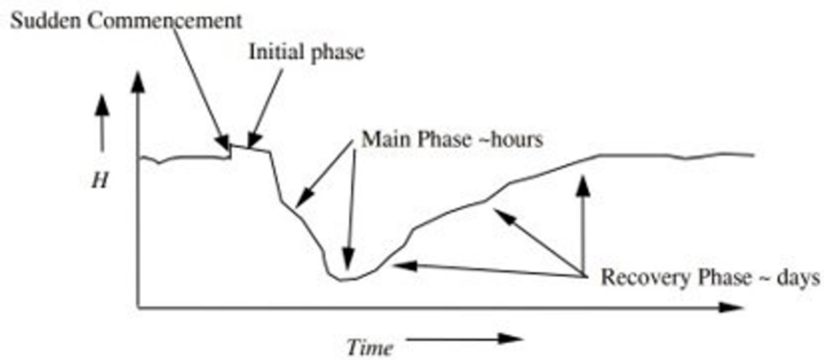

Figure 1: Representation of the stages of a geomagnetic storm

The second possibility is that these charged particles travel along to the charged layers of Earth's magnetic field, travel down the magnetotail, and then eventually "snap back" when magnetic reconnection happens [6]. These particles travel back towards Earth, and due to the anatomy of the magnetic field, come in on the North and South Poles of Earth, causing strong interference in these regions and Aurora Borealis. This phenomenon is called a geomagnetic substorm and is fairly common. As shown in the image below, substorms are much quicker in every phase than a geomagnetic storm, and last only a fraction of the total time.

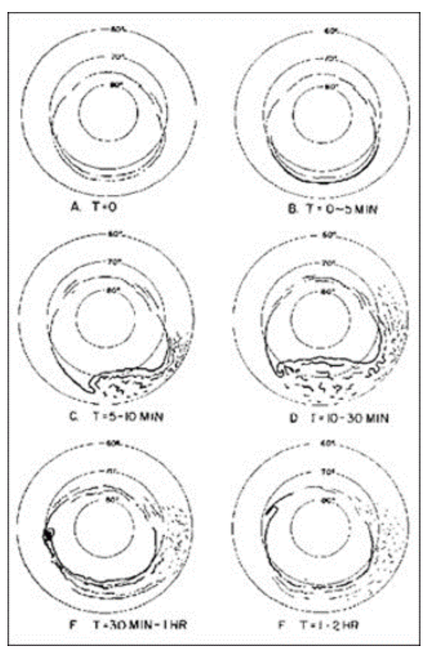

Figure 2: Representation of the stages of a geomagnetic substorm

Geomagnetic substorms can last up to several hours, but typically do not last more than 2 hours. Geomagnetic substorms commonly occur several times per geomagnetic storm, but can occur independently of geomagnetic storms [7] [8] [9]. As mentioned in earlier paragraphs, figure 3 below is an excellent representation of the Earth's magnetic field. The arrows, dots, and X's represent charge flow down, around, and throughout the magnetotail and current sheet regions. Figure 4, which almost looks exaggerated, is a realistic illustration of the sheer size of Earth's actual magnetic field, which can extend far past the moon during strong geomagnetic events.

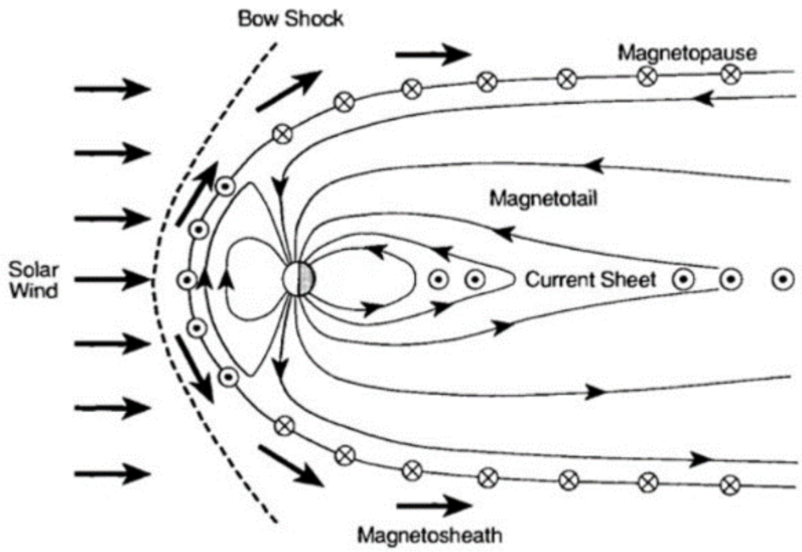

Figure 3: An illustration of the magnetotail, skewed for easier understanding

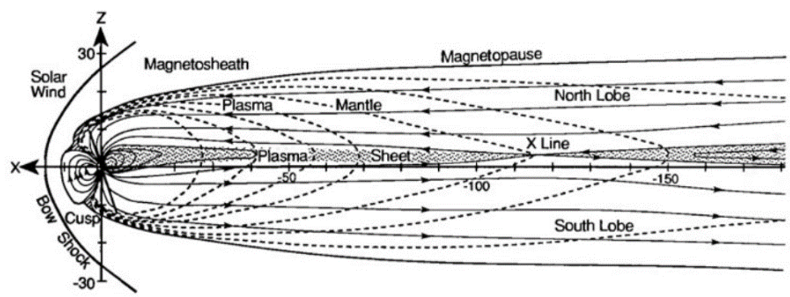

Figure 4: A more realistic illustration of the actual scale of the magnetotail

As for the indices, we will only cover Kp, ap, and Dst. Kp and ap are basically the Richter Scale, but for Geomagnetic Events. Kp and ap have a logarithmic relationship, with ap being the "higher resolution" version of Kp. Kp and ap measure geomagnetic interference, and are the same thing. The only difference is the Kp goes from 0-9 while the ap ranges from 0-400. The Dst measures the geomagnetic interference specifically in the ring current, and any Dst below -50 nT is considered "significant." For scale, a disturbance in the ring current of -50 nT would translate to roughly a G3 class storm, with G5 being your strongest 1-2 per lifetime storms [10] [4]. As observed in figure 5, both graphs represent February of 1986. While the graphs look similar, the top graph, showing the ap, shows much more detail in relation to the geomagnetic event than the bottom graph, representing the Kp.

The idea of Solar Weather Forecasting has been explored before, and has been done somewhat accurately using the Dst index [11]. For this reason, I have selected Kp and ap to see if any new valuable insights can be achieved. While predicting these larger indices can be trivial at times, predicting events such as CMEs will almost always prove challenging [12]. If this can be achieved for Earth, it would also be interesting to see if geospatial orbit technologies (22,300 miles from the

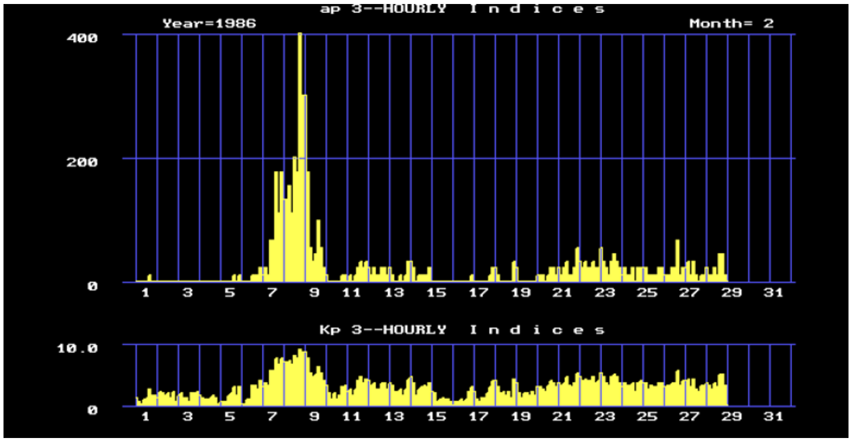

Figure 5: The logarithmic nature of ap (top graph) can be seen when compared to the standard nature of Kp (bottom graph)

equator, located in the furthest orbital region of the Earth in space) provide any different insights, as they tank much more radiation from solar flares and take a beating from CMEs much more often than Earth [13].

3 Data

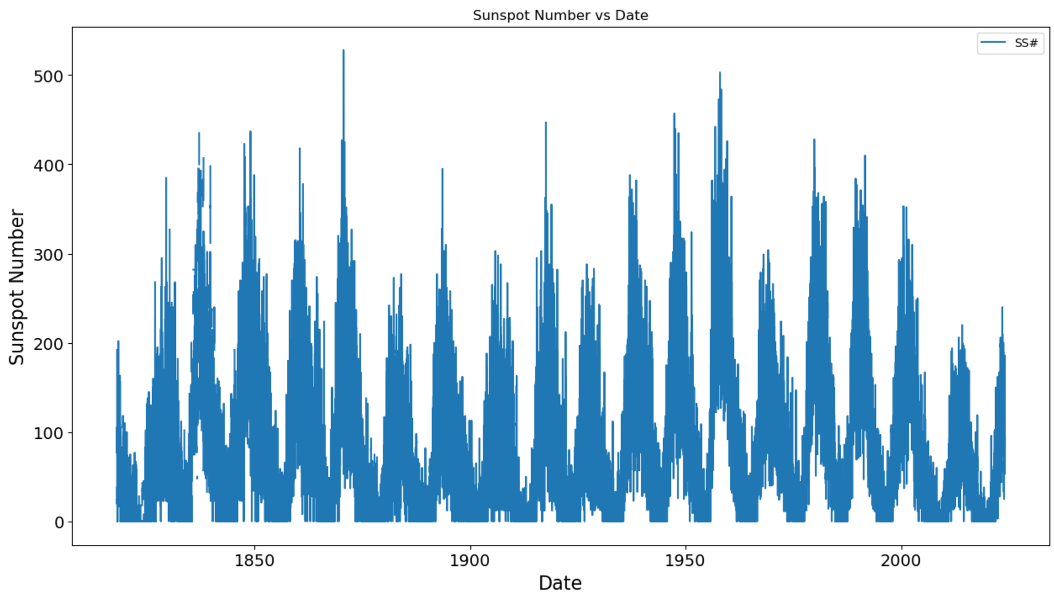

Figure 6: Historical sunspot number versus date from SIDC, graphed

In my analysis of the solar cycle, I used the Solar Influences Data Analysis Center's (SIDC) historical sunspot data. This data set is open source, and is considered a trustworthy source in the solar weather community. The frequency of this data is daily, and has been collected since 01/01/1818. The table contains the calendar date, the fractional date (ex. 1818.001), the daily total sunspot number, the daily standard deviation of sunspot numbers from inputs at different observation stations, the number of observations used to compute the value, and the provisional indicator [2]. The sample size of this data is just over 75,000 observations or rows of data. Of course, as technology has advanced, the number of "null" values in the data has greatly decreased, and the numbers have become more trustworthy. For earlier years that did not have much data, I used linear interpolation in accordance with typical sunspot trends to fill in the data. This provided that every date had a number that would have been close to the real value for that day, even if there were no observations recorded for the date. It is also important to recognize that everything from early years should be taken lightly, as their technology limited their collection methods versus the technology that is observed in today's collection of the same data. This includes factors such as telescope focal length, solar filter quality, counting tools (eyes versus electronic), and observation stations (0-1 for data collected until 1981 versus roughly 70 average in data collected past 1981).

1987	5	27	1987.401	42	2.6	17
1987	5	28	1987.404	25	2.8	18
1987	5	29	1987.407	20	1.9	19
1987	5	30	1987.410	18	1.0	15
1987	5	31	1987.412	14	1.5	21
1987	6	01	1987.415	16	1.5	22
1987	6	02	1987.418	12	3.0	13
1987	6	03	1987.421	0	2.0	18

Figure 7: An example of the data pulled straight from the SIDC csv file

4 Methods

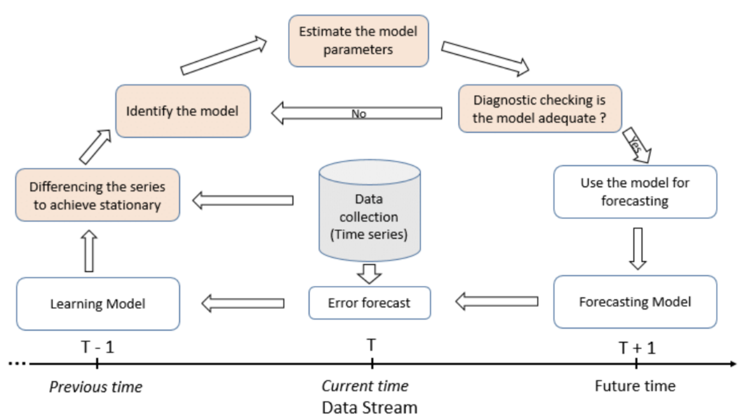

Figure 8: Pipeline of the SARIMAX model

$$y_t = \delta_0 + \delta_1 t + \phi_1 y_{t-1} + _{\dots} + \phi_p y_{t-p} + \Sigma K_j X_{t,j} + \varepsilon_t$$

Figure 9: The SARIMAX equation

The SARIMAX pipeline begins in the middle, with the dataset. Next, differencing is performed, in which the dataset is made stationary. After this, identifying the proper model is needed. For my situation, because I was using data with a seasonality, I used the SARIMAX model. The hardest part is estimating the model parameters, which takes guesswork and graphing aspects such as the acf and pacf of the data. Next, one should visually analyze the data. Is the prediction kind of close? Could we do better? Note that this step either goes to the next step, or goes back to change the parameters.

After an adequate model is obtained, we apply it to future dates, in order to achieve the best prediction of the future as possible.

It should be noted specifically that the SARIMAX model is very data and time intensive, so data may be split into smaller chunks to tackle easier. For example, for my dataset, I split my data into quarterly points, which allowed me to only have to deal with 618 data points but still covered 205 years of observations. Below, the output of running the SARIMAX function can be seen.

Dep. Variable:	y	No. Observations:	618
Model:	SARIMAX(3, 0, 10)x(1, 1, [], 43)	Log Likelihood	-2699.046
Date:	Tue, 02 Apr 2024	AIC	5428.093
Time:	15:36:35	BIC	5493.408
Sample:	03-31-1818 - 06-30-1972	HQIC	5453.567
Covariance Type:	opg		

Figure 10: The output of the SARIMA function that provides details from the algorithm

The SARIMAX model comes from Statsmodel's autoreg library, which contains access to several machine learning algorithms, including ARIMA and other models similar to SARIMAX. This library specializes in forecasting and trending data, with or without seasonality.

For specific implementation see Jack Brewster's Github Page.

5 Results

Thus far, with the best parameters I have found to fit to my SARIMAX model, these are the results that I have for predicting the rest of solar cycle 25.

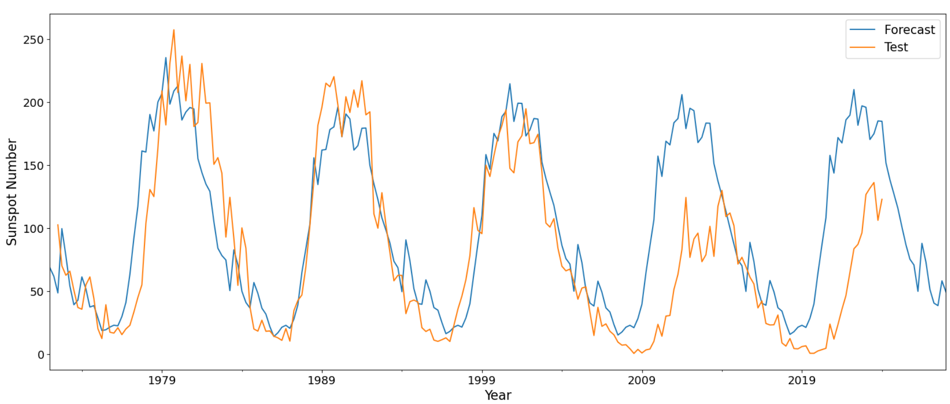

Figure 11: The current SARIMAX fit

While the fit in the model above is not great and could always get better, I am working towards implementing a "semi automatic" SARIMAX, that will be very data and time intensive even more so but will run roughly 100 SARIMAX in sequence, searching for the best fit. This will be similar to a "grid search," and will essentially run the SARIMAX repeatedly, and only keep the best fit. This will run every possible combination for a user specified range of numbers, which will most likely be ±4 around the current fit.

Figure 12 above shows the fit from Figure 11 applied to the remainder of Solar Cycle 25, which is generally expected to reach its next low point around 2029 - 2030. Figure 13 shows the sunspot

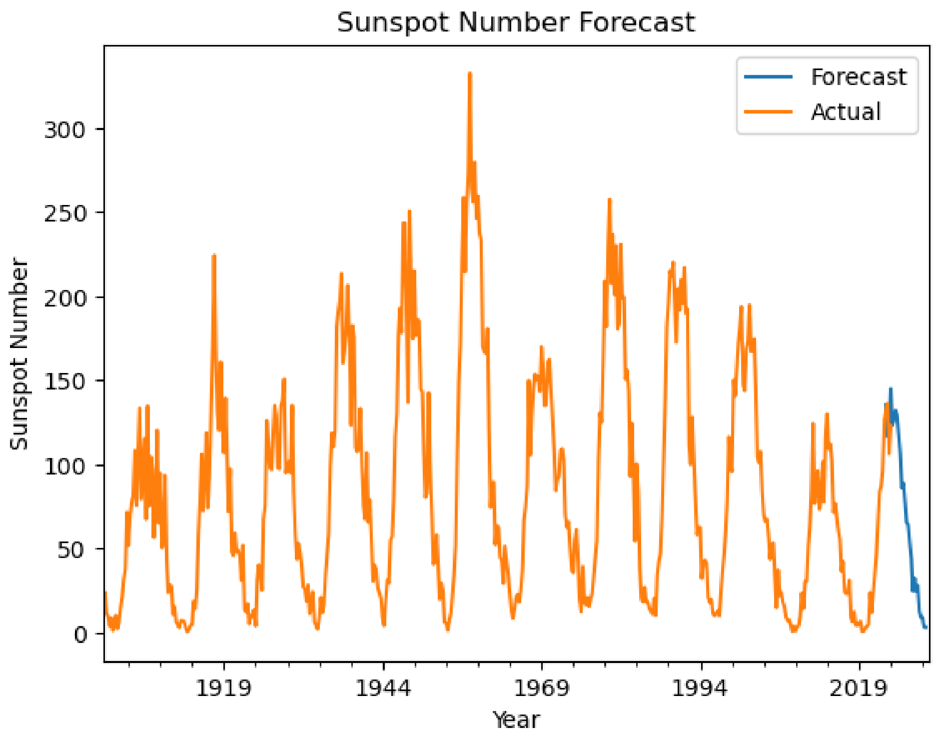

Figure 12: The SARIMAX prediction until 2030 appears on trend

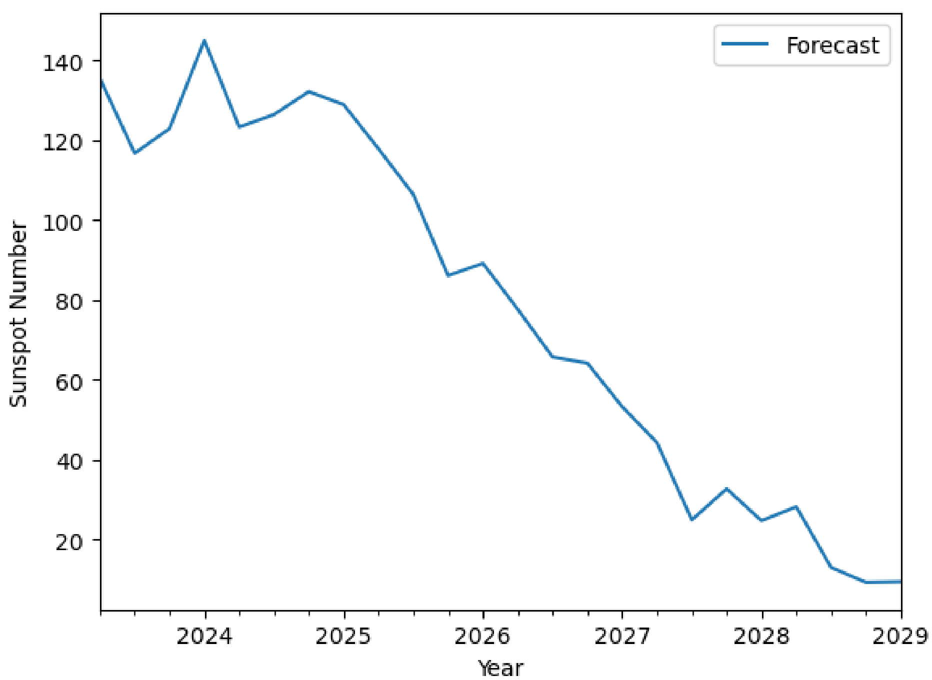

Figure 13: The SARIMAX equation's predictions, 2024-2029

predictions for the remainder of Solar Cycle 25, but should be taken generally. These predictions will be roughly accurate, but have proved unfaithful when predicting in depth and very specific numbers. I hope to improve the fit and achieve better numbers with the "semi automatic" fitting method mentioned previously.

6 Future Work

As far as future work is concerned, I have only met roughly half of my goal thus far. I would not only like to incorporate the Kp and ap into this, but possibly Dst and satellite data as discussed in the data section. For now, I think a reasonable spot to continue the work would be first achieving a better fit, and next incorporating Kp and ap into this. In the graph below, The Kp values from 2020, a historically low activity solar cycle are represented by the blue line versus the Kp from 2001, a recent very active solar cycle represented by the orange line can be observed. From this graph alone, you can obviously see the correlation between the solar cycle and geomagnetic event strength. I believe that further exploring this will provide value.

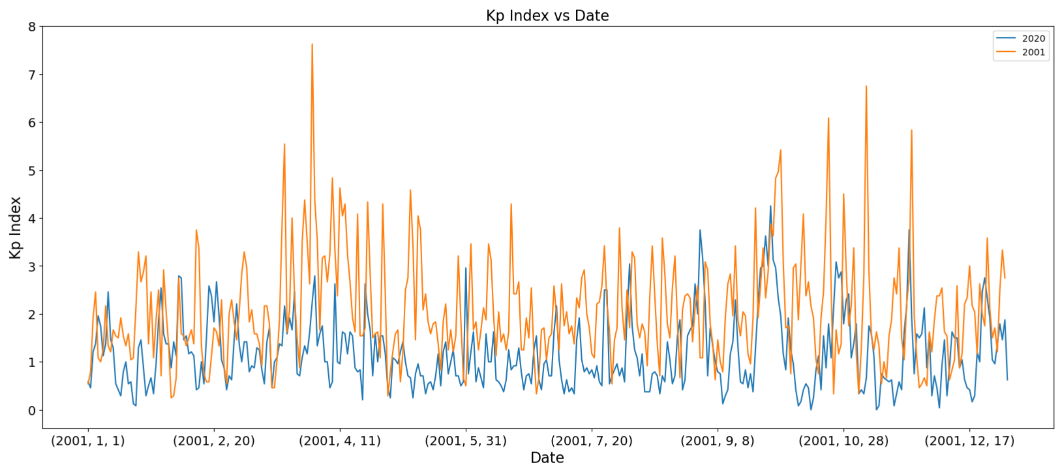

Figure 14: The Kp throughout a solar minimum year versus throughout a solar maximum year

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